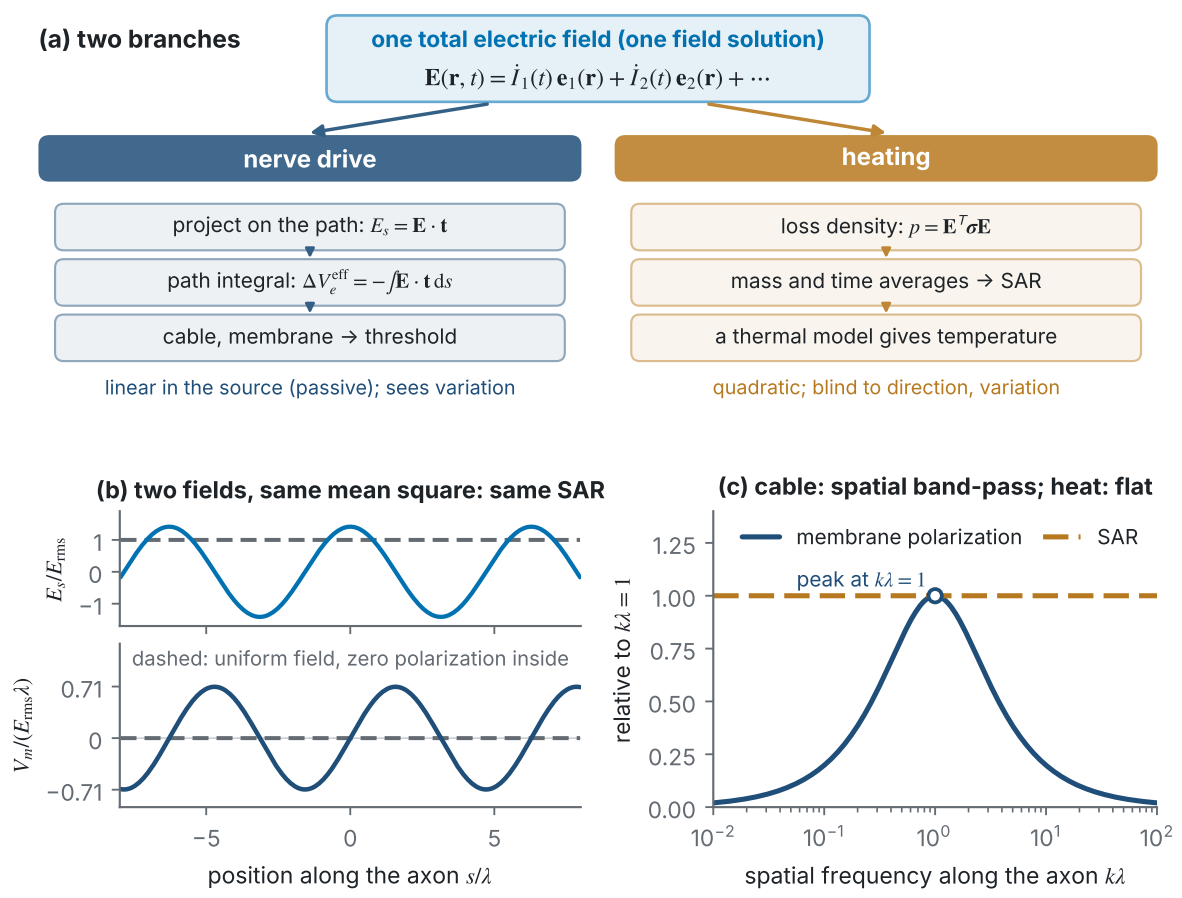
(a) two branches
one total electric field (one field solution)
$\mathbf{E}(\mathbf{r},t) = \dot{I}_1(t)\,\mathbf{e}_1(\mathbf{r}) + \dot{I}_2(t)\,\mathbf{e}_2(\mathbf{r}) + \cdots$
nerve drive
project on the path: $E_s = \mathbf{E}\cdot\mathbf{t}$
path integral: $\Delta V_e^{\text{eff}} = -\int \mathbf{E}\cdot\mathbf{t}\,ds$
cable, membrane → threshold
linear in the source (passive); sees variation
heating
loss density: $p = \mathbf{E}^T\boldsymbol{\sigma}\mathbf{E}$
mass and time averages → SAR
a thermal model gives temperature
quadratic; blind to direction, variation
(b) two fields, same mean square: same SAR
E_s/E_{rms}
1
0
−1
dashed: uniform field, zero polarization inside
$V_m/(E_{\text{rms}}\lambda)$
0.71
0
−0.71
−5
0
5
position along the axon s/λ
(c) cable: spatial band-pass; heat: flat
membrane polarization
SAR
peak at $k\lambda = 1$
relative to $k\lambda = 1$
1.25
1.00
0.75
0.50
0.25
0.00
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
spatial frequency along the axon $k\lambda$